Profile-Driven Hardware Simulation for Organic Optoelectronic Edge Vision

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Abstract

We present a profile-driven behavioral simulation framework for organic optoelectronic compute-in-memory inference that maps literature device metrics to task-level vision accuracy. Across ResNet-18, ConvNeXt-Tiny, and Tiny-ViT-5M on CIFAR-10/100 and Flowers-102, ADC resolution dominates under the canonical regime: below 6 bits causes abrupt collapse, while above 6 bits device-to-device variability becomes binding. Standard hardware-aware training overfits a single fixed mismatch map and collapses to 10.00% on fresh hardware instances, whereas Ensemble HAT—which resamples mismatch masks at each training epoch—recovers 86.16±0.19% across three training seeds. A literature-anchored OPECT case study reaches 88.53±0.08% zero-shot transfer without retraining. Under severe nonlinear write (NL=2.0), hardware-aware training with the revised gradient-scaling recipe recovers to the ∼80–82% band, leaving a residual gap of roughly 15 pp to the digital baseline that reflects a joint effect of non-linear write dynamics and fresh-instance transfer.

Keywords: organic optoelectronic compute-in-memory; hardware-aware training; vision transformers; edge vision; mixed-signal inference; profile-driven simulation.

1 Introduction

Compute-in-memory (CIM) architectures execute dense matrix-vector multiplications directly within memory arrays, eliminating the data-movement bottleneck that limits conventional digital systems[1–3]. Beyond conventional static random-access memory (SRAM) and emerging resistive memories, organic optoelectronic devices offer multilevel tunability, low-voltage operation, and inherent photosensitivity for direct optical signal processing[4–6]. Recent demonstrations span analog conductance states, long retention tails, and integrated optoelectronic synaptic arrays[5;7–11].

However, translating these device-level advances into system-level vision performance faces a critical methodological gap. The organic optoelectronic literature remains dominated by device-centric characterizations[12]. Network-level studies are typically limited to small benchmarks or simple perceptrons, while variability, ADC resolution, retention-induced drift, and transfer across fabricated instances are either absent from system analyses or treated only qualitatively[7;8]. The most relevant precursor for variability-aware organic simulation remains the Monte Carlo study of Lin *et al.*, which sampled device parameters for a simple organic-memristive perceptron[13]. No clear framework yet connects reported organic-device characteristics to deployment-facing accuracy expectations for modern Vision Transformer (ViT) or convolutional neural network (CNN) backbones.

Here we present a behavioral simulation framework that bridges this materials-to-system gap, providing structured identification and ranking of hardware-induced failure modes—what we call risk-aware evaluation—prior to fabrication closure. The framework incorporates organic-device-specific non-idealities—photoresponse, retention, and write nonlinearity—through a replaceable device-profile interface and applies a mixed-signal deployment model to ResNet-18, ConvNeXt-Tiny, and Tiny-ViT-5M. Four concrete contributions follow. First, a profile-driven behavioral workflow for organic optoelectronic CIM that maps literature-derived device characteristics to task-level vision accuracy across CNN and ViT backbones. Second, identification of dominant deployment constraints through a sub-6-bit ADC cliff ($S_{\text{ADC}} = 0.98$) and an inverse-gamma front-end compensation that restores accuracy for strongly sublinear photoresponse ($\gamma_{\text{phys}} = 2.0$, $+5.8$ pp) while trading dark-region linearity for bright-region shot-noise amplification. Third, exposure of a fresh-instance transfer failure in standard HAT and introduction of Ensemble HAT, which resamples D2D masks each epoch to raise fresh-instance accuracy from 10.00% to 86.16±0.19% across three training seeds. Fourth, zero-shot transfer to a literature-anchored OPECT profile and localization of the $NL = 2.0$ surrogate failure primarily to the MLP path.

The framework is intentionally behavioral rather than circuit-accurate: array-level parasitics (current–resistance, or IR, drop; sneak paths), thermal effects, and pulse-faithful programming dynamics are not modeled explicitly and are revisited in Section 4. Its purpose is to identify which device characteristics most constrain edge-vision accuracy before full array hardware becomes available, and to provide a reproducible workflow for comparing mitigation strategies under a common deployment model.

2 Related Work

2.1 Hardware-Aware Training and Robustness

Hardware-aware training (HAT) mitigates analog non-idealities by injecting hardware perturbations into the forward path during optimization[14]. Although related in spirit to quantization-aware training[15] and to domain randomization in sim-to-real transfer[16], HAT for CIM faces an additional difficulty: device-to-device (D2D) mismatch is spatially structured and fixed for a given hardware instance. It is therefore different from independent thermal or sampling noise. Ensemble HAT resamples the static D2D mask at each training epoch, improving zero-shot transfer to fresh hardware instances. The connection to domain randomization[16] is deliberate, but the analogy is superficial because organic optoelectronic CIM exhibits spatially structured D2D mismatch, C2C noise, and nonlinear write that are fixed per hardware instance.

2.2 Organic Neuromorphic and Optoelectronic Devices

Organic synaptic and memristive devices offer optical sensitivity, multilevel conductance tuning, and flexible-substrate compatibility for neuromorphic hardware[4;6;17;18]. Recent demonstrations include multilevel memory[5;7], long retention tails[8], and integrated optoelectronic arrays[9;10;19;20]. Array-level studies are beginning to treat active-matrix addressing, spatial optical response, and crosstalk as system constraints rather than purely materials observations[21–23], while temperature-resilient organic synapses and high-temperature synaptic phototransistors suggest that thermal stability is also becoming deployment-relevant[24;25].

2.3 CIM Simulation Frameworks and Hybrid Mapping

System-level CIM simulators such as DNN+NeuroSim established an effective template for connecting device assumptions, peripheral-circuit costs, and neural-network accuracy[2]. MemTorch showed how memristive non-idealities can be embedded into PyTorch workflows for end-to-end evaluation[26], AIHWKIT exposed analog HAT and inference simulation in a practical mixed-signal software stack[27], and CrossSim provides a GPU-accelerated crossbar-accuracy workflow with parasitic resistance, ADC, drift, and lookup-table-based training models[28]. These mature simulators were developed primarily around inorganic resistive memories, and they do not directly expose inverse-gamma optoelectronic photoresponse or organic-specific double-exponential retention as first-class device primitives. Our behavioral profile-driven approach therefore serves as an organic-specific complement rather than as a direct replacement for these inorganic toolkits.

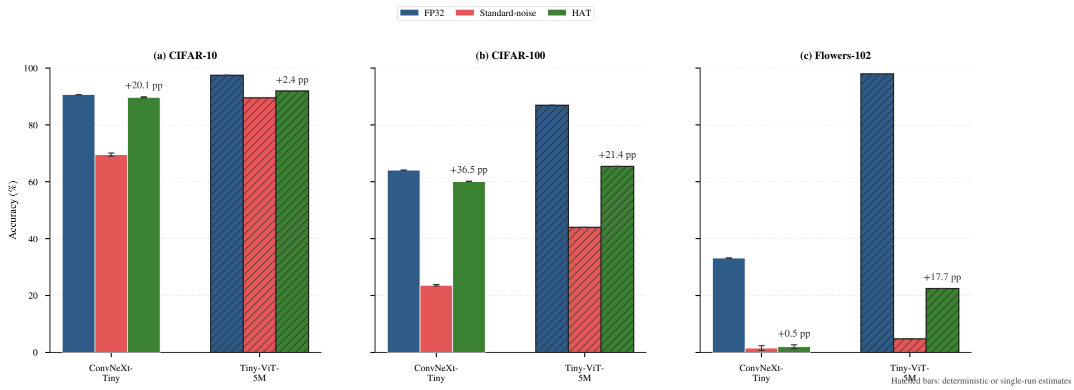

Figure 1: Cross-dataset accuracy under canonical deployment (4-bit quantization, 5% C2C, 10% D2D variability) for Tiny-ViT-5M. Error bars denote ±1 standard deviation across Monte Carlo (MC) forward-pass averages ($n = 10$ fresh instances, 5 runs per instance); bars without visible error bars indicate deterministic baselines (FP32 and V2 zero-noise hybrid).

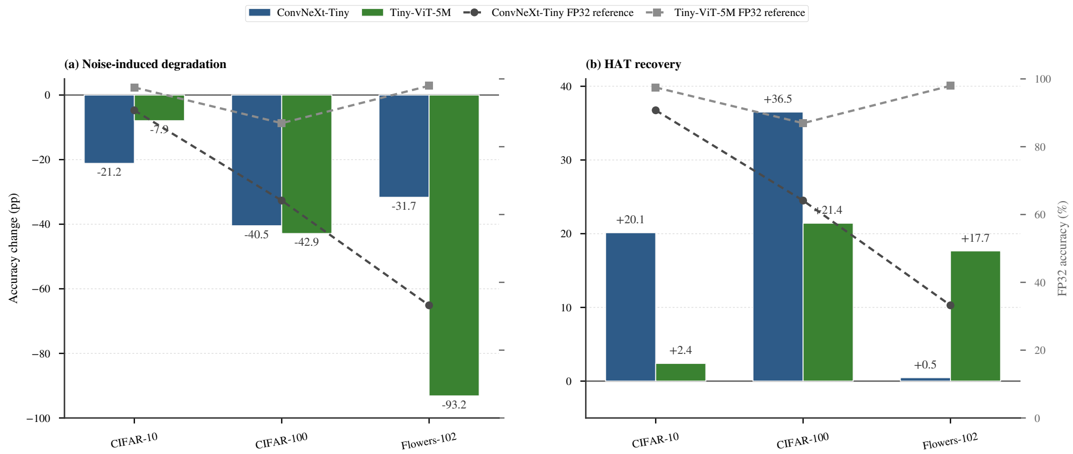

Figure 2: Accuracy degradation from FP32 to noisy deployment and HAT recovery on CIFAR-10. Tiny-ViT: 10 fresh-instance MC averages (5 runs per instance); ConvNeXt: single-run deterministic estimates.

3 Results

3.1 Baseline Digital Performance

Digital FP32 baselines are 95.46% (ResNet-18), 90.74% (ConvNeXt-Tiny), and 98.06% (Tiny-ViT-5M) on CIFAR-10, with corresponding CIFAR-100 values of 78.64%, 64.12%, and 86.94%.

3.2 Quantization and Noise Resilience

The zero-noise hybrid control (V2; Table 2) retains 97.39±0.00% ($n = 10$ deterministic evaluations). The recovered-weight law (Eq. 1) suppresses much effective analog perturbation below the 4-bit LSB.

On CIFAR-100, HAT improves Tiny-ViT from 44.06% to 65.48% and ConvNeXt from 23.86% to 60.54%.

The same 6-bit threshold holds on ConvNeXt-Tiny (Supplementary Fig. S3), and the iso-accuracy contour analysis (Section 3.6) shows this cliff is independent of device variability, producing a consistent ∼7 pp transition across all tested σ_{D2D} levels.

3.3 Retention and Temporal Drift

The corrected retention-aware follow-up model (V8) exhibits a rapid early drop (91.63% at 0 s $\rightarrow$ 82.66% at 1 s) followed by a broad plateau near 79% (79.13–79.51%) through 10 000 s (Supplementary Fig. S6).

3.4 Hardware-Instance Transferability

Standard fixed-mask HAT (V4) collapses to chance level (10.00%) on arrays with different fixed D2D realizations, indicating pronounced hardware-instance overfitting. The collapse was independently confirmed under no-AMP FP32 inference (Supplementary Note S-Verification). This motivates the distribution-matched Ensemble HAT objective (Eq. 3), formally defined in Section 5 and instantiated in Section 3.7.

A canonical class-distribution diagnostic clarifies the collapse mechanism. Re-evaluating the fixed-mask Standard HAT checkpoint on five fresh D2D instances yields $10.00 \pm 0.00\%$ accuracy; for every instance, all 10 000 test images are assigned to a single CIFAR-10 class (prediction entropy ≈ 0, maximum-class frequency $= 100\%$). Under the same five-instance protocol, canonical Ensemble HAT reaches $85.97 \pm 1.98\%$ with near-uniform prediction entropy (2.28 vs. $\ln 10$) and a maximum-class frequency of $15.27 \pm 2.27\%$ (Supplementary Fig. S20). Thus, the fixed-mask failure is a deterministic single-class hardware-instance collapse rather than Monte Carlo noise or mixed-precision evaluation drift, and epoch-level D2D resampling prevents this degenerate predictor.

3.5 Physical Front-end Compensation

The organic phototransistor front end exhibits a sublinear photoresponse $I_{\text{photo}} \propto P^{\gamma_{\text{phys}}}$ that we compensate with an inverse-gamma preprocessor $P_{\text{in}} = X^{1/\gamma_{\text{phys}}}$ (Eq. 5). Compensation restores **+5.8 pp** at $\gamma_{\text{phys}} = 2.0$ (89.85% vs. 84.04%, single-seed deterministic evaluation), but amplifies shot-noise variance for $\gamma_{\text{phys}} < 1$. The full γ_{phys} scan and the analytical trade-off are shown in Supplementary Figs. S7–S8.

3.6 Iso-Accuracy Operating Envelope

A 63-point joint sweep over $\sigma_{\text{D2D}} \in \{1, 3, 5, 8, 10, 15, 20\}\%$ and ADC $\in \{2, 3, 4, 5, 6, 7, 8, 10, 12\}$ bits (10 MC runs per point) reveals three regimes (Figure 3). Below 5-bit ADC, accuracy collapses to near-chance; the 5-to-6-bit transition yields a consistent $\sim$7 pp jump. Above 6 bits, degradation is dominated by D2D: $\leq$10% D2D maintains >84%, while 20% D2D reduces it to 71–77%. Sobol indices (Eq. 8) are $S_{\text{ADC}} = 0.98$ over the full grid and $S_{\text{D2D}} = 0.92$ in the operational region ($\geq$6 bits, $\sigma_{\text{D2D}} \leq 15\%$), with negligible interaction (<4%).

3.7 Nonlinear Writing and Hardware-Aware Training

Proportional-noise HAT recovers Tiny-ViT to 97.37±0.05% when training and evaluation physics are aligned, but falls to 10.38±0.44% under uniform-noise semantics. For ConvNeXt-Tiny, proportional-noise HAT reaches 91.98% (91.91±0.08% Monte Carlo).

Ensemble HAT (Figure 4; Eq. 3) preserves 86.16±0.19% across three protocol-matched training seeds under canonical uniform noise ($NL = 1.0$), with each seed evaluated on 10 fresh arrays and five Monte Carlo passes per array. The original plotted canonical checkpoint remains 86.37±1.54% across its 10 fresh arrays, and all three seeds remain far above standard fixed-mask HAT ($10.00 \pm 0.00\%$). An ablation scan (Supplementary Fig. S4) shows epoch-level resampling reaches 88.41% (50-epoch), versus 87.18% (fixed) and 86.16% (per-batch), confirming structured epoch-level resampling as the load-bearing schedule choice.

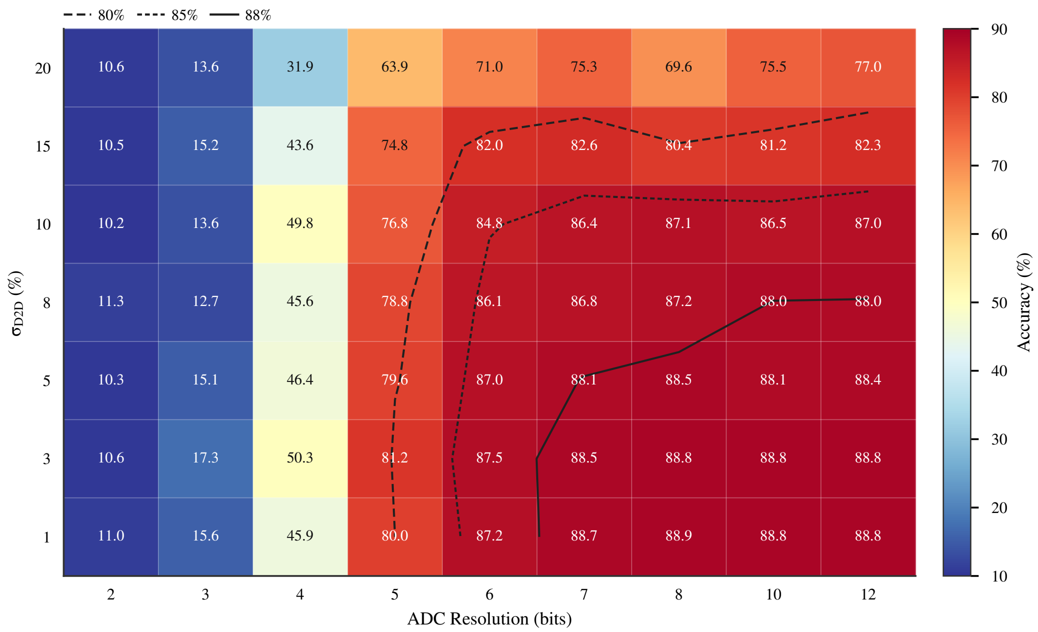

Figure 3: Iso-accuracy contour map for the Ensemble HAT Tiny-ViT model under joint σ_{D2D} and ADC precision sweep ($\sigma_{\mathrm{C2C}} = 5\%$, $NL = 1.0$, 10 MC runs per grid point). The 6-bit ADC cliff separates the collapse regime (below 5 bits, near-chance accuracy) from the operational regime (above 6 bits, D2D-dominated degradation).

Severe-NL retraining after gradient-scaling correction. Retraining under severe write non-linearity ($NL = 2.0$) was repeated with a revised gradient-scaling recipe (Section 5.2). Table 1 summarizes the local M-series results; remote batch-512 results (83.64% Standard, 84.80% Proportional) are provided in the Supplementary Information. All packets use true $NL_{\mathrm{LTP}} = 2.0$ and $NL_{\mathrm{LTD}} = -2.0$ in both training and evaluation. Per-instance ADC range recalibration confirms the static-cal protocol does not bias the headline.

Table 1: Severe Nonlinear Writing Recovery (M-Series). All models trained and evaluated with true $NL_{\mathrm{LTP}} = 2.0$ and $NL_{\mathrm{LTD}} = -2.0$. Headline numbers are post-module-output hook diagnostic 8-bit ADC quantization with per-instance range recalibration on each fresh hardware realization (ADC-on); ADC-off training-surrogate baselines differ by approximately -0.10 pp on average (see text). Remote batch-512 results (83.64% Standard, 84.80% Proportional) are provided in the Supplementary Information. The ADC-on values are diagnostic-only and should not be treated as deployment-fidelity claims until a physical ADC boundary is implemented.

Task	HAT Variant	Noise	Seed	ADC-on 8-bit (%)	Std (%)	ADC-off baseline (%)
M1	Standard (V3)	Uniform	123	81.89	1.02	82.03
M5	Standard (V3)	Uniform	456	80.37	0.08	80.47
M2	Ensemble (V4)	Uniform	123	80.37	0.59	80.45
M6	Ensemble (V4)	Uniform	456	81.04	1.73	81.18
M3	Ensemble (V4)	Proportional	123	80.64	0.13	80.71
M4	Ensemble (V4)	Proportional	456	80.67	0.41	80.75

An intermediate NL sweep (Ensemble HAT, seed 123, uniform noise) bridges the canonical and severe regimes. At $NL = 1.2$ the fresh-instance mean is $83.03 \pm 0.22\%$; at $NL = 1.5$ it is $82.63 \pm 0.10\%$; and at $NL = 1.8$ it falls to $80.31 \pm 0.40\%$ (Supplementary Table **??**). The degradation is gradual rather than abrupt, with the largest drop occurring between $NL = 1.5$ and $NL = 1.8$. This monotonic trend supports the interpretation that the $\sim$5 pp residual gap at $NL = 2.0$ reflects a training-recipe limitation in the strongly nonlinear regime, not a physical accuracy floor.

Four observations emerge from the M-series evidence (seeds 123, 456, 789). First, all true-NL2

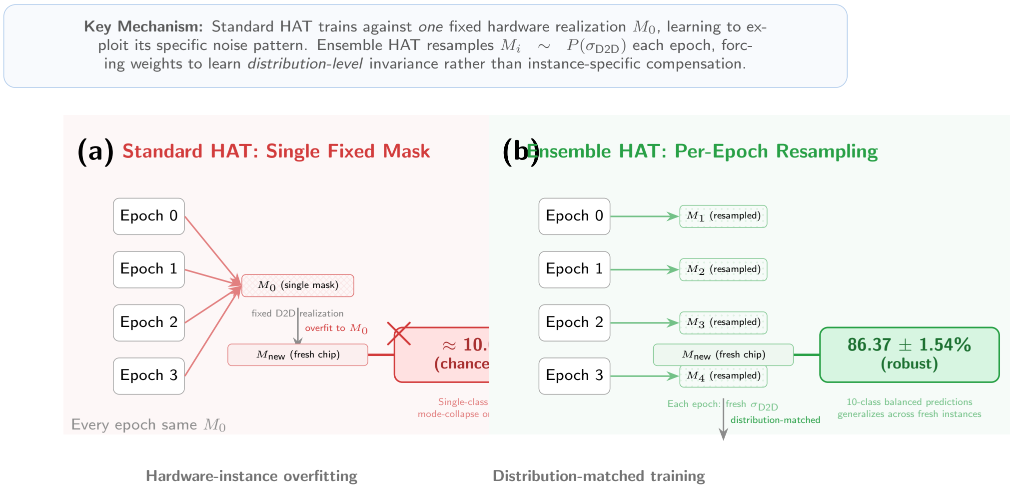

Figure 4: Ensemble HAT concept. Standard HAT reuses one fixed D2D mismatch map throughout training, whereas Ensemble HAT resamples the mismatch map at each epoch. The fresh-instance transfer panel (right) shows the original canonical checkpoint: standard HAT collapses to chance level on unseen hardware instances, while Ensemble HAT maintains $86.37 \pm 1.54\%$ across ten fresh arrays. A protocol-matched three-seed rerun gives $86.16 \pm 0.19\%$ across seed-level fresh-instance means.

routes recover to ∼80–82%, irrespective of noise law or D2D resampling. Second, Standard and Ensemble HAT do not separate within seed variance (Standard uniform 81.20% vs Ensemble uniform 80.54%). Third, proportional noise offers no advantage under $NL = 2.0$ (80.83% vs uniform 80.54%). Fourth, 8-bit ADC quantization does not materially alter the headline (−0.10 pp), while a 6-bit spot-check reveals a −2.8 pp cliff, consistent with §3.6.

Severe-NL retraining recovers to ∼80–82%, leaving a residual gap of roughly 15 pp to the digital baseline.

3.8 Case Study: Zero-Shot Transfer to a Literature-Calibrated Device

We simulated zero-shot transfer to a 2025 OPECT array[10] ($\sigma_{\mathrm{C2C}} = 2\%, \sigma_{\mathrm{D2D}} = 3\%$). Figure 5 shows that standard-HAT checkpoints transfer poorly: ConvNeXt retains partial transfer, whereas Tiny-ViT V4 collapses to chance level (10.00%) across all alternatives. By contrast, the Ensemble-HAT checkpoint reaches **88.53±0.08%** ($n = 10$ fresh-instance evaluations) under the nominal OPECT proxy.

The OPECT case study[10] provides **one literature-anchored validation point**; we choose it because it is the most complete published organic-array characterization to date, with documented D2D, C2C, and conductance-window statistics. The framework is **profile-agnostic by design**: any measured device statistics can be ingested through the documented profile interface (Supplementary Note S-HW). We anticipate cross-validation against additional fabricated profiles in a companion study; the present zero-shot transfer demonstrates the framework's portability without retraining, not a claim that OPECT is universally representative.

4 Discussion

4.1 Diagnosis: hardware-instance overfitting as the primary failure mode

Once hardware-aware training (HAT) and scale recovery are available, standard cycle-to-cycle (C2C) and device-to-device (D2D) variability do not dominate the error budget. Three constraints dominate instead.

A quantitative decomposition supports this ordering: designers should first secure 6-bit readout, then invest the remaining error budget in reducing device-to-device mismatch.

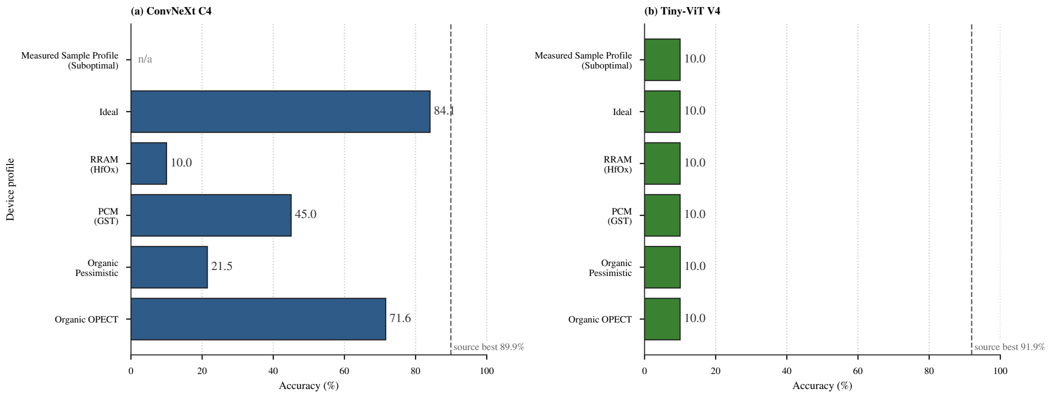

Figure 5: Zero-shot transfer across alternative device profiles. Standard-HAT checkpoints are evaluated on literature-calibrated or measured alternative profiles without profile-specific retraining. ConvNeXt C4 retains partial transfer on OPECT and idealized profiles but degrades on PCM and RRAM; Tiny-ViT V4 collapses to chance level (10.00%) across all alternatives. By contrast, Ensemble-HAT reaches $88.53 \pm 0.08\%$ on the OPECT proxy. Dashed vertical lines mark source-profile best accuracy; n/a indicates unavailable combinations.

The first is ADC resolution. Below 6 bits, transformer inference degrades substantially, indicating that conductance control remains gated by readout precision.

The second is hardware-instance overfitting. The collapse of V4 under fresh-instance transfer (10.00%) shows that standard HAT readily fits a single fixed D2D realization. Replacing the fixed-mask objective (Eq. 2) with the mismatch-distributed objective (Eq. 3) raises fresh-instance accuracy to $86.16 \pm 0.19\%$ across three training seeds. Mechanistically, Ensemble HAT changes the mismatch map across epochs rather than merely perturbing one nominal instance.

A residual gap to the 98.06% digital baseline persists; part may reflect regularization from per-epoch D2D resampling. Capacity-aware Ensemble HAT schedules remain an open problem.

The third is the scale-masking regime in V2. When the recovered-weight law (Eq. 1) maps analog perturbations below the 4-bit quantization step, many realizations remain in the same recovered weight bin. This protection disappears under state-dependent noise.

4.2 Treatment: Ensemble HAT mitigation across scenarios

4.3 Mechanism: theory and empirical landscape

Supplementary Note S-Theory derives that the per-epoch resampling objective is, to second order, equivalent to a weighted gradient-L_2 regularizer (analogous to dropout-as-L_2, Wager et al. 2013) and, structurally, to a stochastic Sharpness-Aware Minimization along the device-mismatch direction (analogous to SAM, Foret et al. 2021). A PAC-Bayes argument provides a generalization-bound rationale for why fresh hardware-instance accuracy tracks the training-distribution average.

Empirical diagnostics support this interpretation. Loss-landscape interpolation from the training D2D mask toward fresh masks (Supplementary Note S-Mechanism, E2) shows that Ensemble HAT maintains 88.39% accuracy when evaluated on a fresh mismatch map, whereas Standard HAT collapses to 10.00%—a gap of 78.39 pp. Under extrapolated mismatch ($3\times$ amplitude), Ensemble HAT still outperforms Standard HAT by 17.5 pp, indicating that the robustness extends beyond the training distribution. Per-layer sensitivity analysis (Supplementary Note S-Mechanism, E4) localizes the analog bottleneck to MLP layers (four of the five most sensitive layers), consistent with the main-text severe-NL narrative.

Full-parameter-space Hessian spectra (Supplementary Note S-Mechanism, E1) reveal a more nuanced picture: Standard HAT exhibits lower top eigenvalues at its training mask than Ensemble HAT, yet this "flatness" is mask-specific and does not transfer to fresh instances. The relevant robustness metric is therefore D2D-directional sensitivity rather than global curvature, aligning with the theoretical regularizer derived in Supplementary Note S-Theory.

4.4 Implications: design rules for deployment

Box 1: Design rules for organic optoelectronic CIM transformer deployment

1. **Secure ≥6-bit ADC readout first.** Below 6-bit, transformer inference degrades by ∼7 pp per row across all tested D2D levels (Section 3.6).
2. **Then minimize device-to-device variability.** Within the operational envelope (≥6-bit, $\sigma_{\mathrm{D2D}} \leq$ 15%), D2D explains 92.2% of residual accuracy variance.
3. **Train with mismatch-distributed objective (Ensemble HAT), not fixed-mask.** Standard fixed-mask HAT collapses on fresh hardware instances (10.00%); Ensemble HAT recovers $86.16 \pm 0.19\%$ canonical across three seeds and $88.53 \pm 0.08\%$ on the literature-calibrated reference.
4. **Per-epoch D2D resampling, not per-batch.** Empirical ablation: epoch-level reaches 88.41%; per-batch only 86.16%; fixed 87.18%.
5. **Apply scale-recovery calibration after readout.** Without it, the analog perturbation is no longer bounded below the LSB and the recovered-weight protection vanishes.
6. **Compensate front-end sublinear photoresponse via inverse-gamma preprocessing.** Restores up to +5.8 pp at $\gamma_{\mathrm{phys}} = 2.0$.
7. **Calibrate against measured D2D distribution when available.** Literature priors are an upper-bound proxy; deployment risk should be re-assessed once fabricated array statistics are in hand (see Supp Note S-HW).

The design rules in Box 1 are intentionally conservative: each threshold is chosen at the point where the accuracy–cost trade-off begins to degrade rapidly, not at the theoretical optimum. The 6-bit ADC threshold is the *cliff* observed in the iso-accuracy sweep, not the minimum ADC that could work under ideal calibration. The 15% D2D ceiling marks the boundary beyond which rank ordering between HAT variants destabilizes. Taken together, the rules frame deployment as sequential thresholding—readout first, then mismatch, then objective.

4.5 Limitations and outlook

Scope and limitations. The conclusions above should be read against the boundaries of the present simulation stack. All experiments rest on device profiles calibrated from literature values or a single fabrication batch, so the accuracy projections do not extend to cross-batch variability, temperature drift, or aging. The severity of analog-induced degradation also scales with task complexity and data scarcity, which remains a fundamental limitation.

The framework is intentionally **behavioral, not circuit-accurate**: we simulate deployment-relevant non-idealities (quantization, D2D/C2C variability, ADC precision, write nonlinearity, retention) but defer SPICE-level phenomena (parasitic IR drop, sneak paths, pulse-faithful programming) to follow-on work. This scope choice mirrors established analog-CIM tooling[27], where behavioral fidelity suffices to **rank deployment risks** pre-fabrication. The profile-agnostic ingest interface accepts any measured device statistics (Supplementary Note S-HW); hardware-calibrated re-evaluation is anticipated once partner-array measurements are released.

The energy estimates are **first-order analytical projections** derived from literature-proxy per-operation constants ($E_{\mathrm{cell}} = 100$ fJ, $E_{\mathrm{ADC},8b} = 25$ fJ, $E_{\mathrm{DAC},8b} = 30$ fJ; Section 5.5). They bound deployment risk to within an order of magnitude rather than predicting silicon performance. We deliberately avoid silicon-grade energy claims because the framework's contribution is **risk-ranking under realistic device profiles**, not chip-level optimization. A complete energy validation depends on partner-array characterization not yet available.

The present evaluation focuses on CIFAR-10/100 and Flowers-102 to enable systematic non-ideality and HAT-method ablation across three backbones within a tractable compute budget. **ImageNet-scale extension is the subject of an in-progress cross-architecture sweep** (ViT-Small/16, DeiT-Small/16 on TinyImageNet) on a separate compute infrastructure; preliminary results will appear in an extended companion study or supplementary erratum. The ablation matrix presented here intentionally trades dataset breadth for non-ideality coverage, providing the structural Sobol decomposition (Section 3.6) that motivates dataset-scale priorities for hardware validation.

Outlook. Three extensions follow naturally: measured-device calibration, cross-architecture validation on ViT-Small and DeiT-Small, and higher-order gradient corrections for the $NL = 2.0$ surrogate. The residual gap between severe-nonlinear-write fresh-instance accuracy and the canonical baseline reflects **training-recipe constraints rather than a physical floor**. The revised gradient-scaling recipe already recovers the ∼80–82% band under $NL = 2.0$ (Table 1), consistent with optimizer-state tuning rather than fundamental robustness limits. We frame the severe-NL regime as an **upper-bound stress test**, not a target deployment specification; whether physical organic devices enter this regime awaits direct nonlinear-write measurements.

5 Methodology

5.1 Hybrid Analog/Digital Deployment

We adopt a hybrid analog/digital inference stack: dense linear operators execute on differential-pair crossbars; control-heavy operators remain digital. Concretely, the analog partition contains patch embeddings, query/key/value (Q/K/V) and output projections, and feed-forward layers; the digital partition contains softmax, layer normalization, GELU activation, positional encoding, residual additions, and the class token, following established practice in heterogeneous CIM accelerators[29–33].

Under this mapping, approximately 88% of parameters execute on analog arrays while roughly 60% of estimated energy remains digital, largely because attention operations stay in the digital domain—a pattern consistent with recent heterogeneous CIM accelerators for Vision Transformers[30–33].

5.2 Modeling Physical Non-Idealities

The framework injects quantization, cycle-to-cycle (C2C) and device-to-device (D2D) variability, ADC distortion, retention drift, and nonlinear-write surrogates directly into the forward path. Each analog-mapped weight is decomposed into positive and negative branches, mapped to the conductance window $[G_{\min}, G_{\max}]$, and quantized to n_{states} levels via a straight-through estimator (STE) quantizer $Q_n(\cdot)$. Differential readout recovers $G_{\text{eff}} = \tilde{G}^{+} - \tilde{G}^{-}$, and the downstream digital weight is

$$\hat{W}_{ij} = s_\ell \, G_{\text{eff},ij}, \qquad s_\ell = \begin{cases} \dfrac{w_{\max}}{G_{\max} - G_{\min}}, & \text{standard calibration} \\[2ex] \dfrac{w_{\max}}{\max_{i,j} |G_{\text{eff},ij}| + \epsilon}, & \text{retention-time recalibration} \end{cases} \tag{1}$$

with layer index ℓ omitted. Unless otherwise stated, we use the standard-calibration branch; the retention-recalibration branch is used only for the V8 retention-drift experiments. The recovered weight step is $\Delta_{\hat{W}} = s_\ell(G_{\max} - G_{\min})/(n_{\text{states}} - 1)$, so perturbations smaller than half a step are masked. The differential-pair scheme suppresses common-mode noise[34;35].

Hardware-aware training (HAT) injects forward perturbations while back-propagating through a straight-through estimator. Standard HAT fixes one D2D mismatch field M_0,

$$\theta^{\star}_{\text{std}} = \arg\min_{\theta} \frac{1}{N} \sum_{n=1}^{N} \ell(f(x_n; \theta, M_0), y_n)\,, \tag{2}$$

Ensemble HAT as distribution matching. Ensemble HAT treats the mismatch map as a random variable resampled each epoch:

$$\theta^{\star}_{\text{ens}} = \arg\min_{\theta} \; \mathbb{E}_{M\sim\mathcal{D}_{\text{D2D}}}\Big[\frac{1}{N}\sum_{n=1}^{N} \ell\big(f(x_n;\theta,M), y_n\big)\Big], \tag{3}$$

where the expectation is Monte-Carlo estimated with one fresh draw $M^{(e)} \sim \mathcal{D}_{\text{D2D}}$ held constant across all mini-batches in epoch e. Expanding Eq. 3 (Supplementary Note S-Theory) reveals an implicit Fisher-weighted gradient-squared regularizer that pushes the optimizer toward flat regions in the M-direction, explaining the fresh-instance transferability without extra hyperparameters. Epoch-level (not per-batch) resampling is load-bearing, giving the optimizer enough consistent signal to fit each instance before resampling[16]. Fresh-instance transfer evaluates on unseen draws $M' \sim p(M)$:

$$\text{Acc}_{\text{fresh}} = \mathbb{E}_{M'\sim p(M)}\big[\text{Acc}\big(f(\cdot;\theta^{\star}_{\text{ens}}, M')\big)\big]\,. \tag{4}$$

Each fresh array is an i.i.d. Gaussian draw $M' \sim \mathcal{N}(0, \sigma^2_{\text{D2D}})$ held constant over the evaluation pass, distinct from per-forward-pass C2C noise. For a single checkpoint, the reported fresh-instance spread is the standard deviation across ten fresh-instance means (each averaged over five forward passes); for the R10A reproducibility headline, the uncertainty is the sample standard deviation across three seed-level fresh-instance means.

5.3 Training Protocol

All Tiny-ViT experiments use AdamW ($lr = 5 \times 10^{-4}$, weight decay 0.05), 100 epochs, batch size 64, cosine annealing ($T_{\max} = 100$). Images are resized to 224×224 with standard augmentation. The default analog resolution is $n_{\text{states}} = 16$ levels. V1 is digital; V2 is hybrid without noise; V3 is hybrid with fixed D2D ($\sigma_{\text{D2D}} = 10\%$); V4–V7 use full HAT with epoch-resampled D2D and per-forward C2C ($\sigma_{\text{C2C}} = 5\%$ or 10%). V6 adds the photoresponse front end; V7 evaluates retention drift.

Nonlinear-write and retention surrogates are anchored to measured DNTT transients[36]. Nonlinear write modifies the STE backward pass by scaling the straight-through gradient with a state-dependent factor proportional to the present conductance magnitude raised to $NL - 1$[1]; ideal write is recovered when $NL_{\text{LTP}} = 1$ (long-term potentiation) and $|NL_{\text{LTD}}| = 1$ (long-term depression). For optoelectronic sensing, we incorporate a physical front-end model (V6) that first applies inverse-gamma preprocessing to the normalized input $X \in [0, 1]$,

$$P_{\text{in}} = X^{1/\gamma_{\text{phys}}}, \tag{5}$$

and then propagates it through a photocurrent response model,

$$I_{\text{photo}} = \alpha P_{\text{in}}^{\gamma_{\text{phys}}} + I_{\text{dark}} + \varepsilon_{\text{shot}} \approx \alpha X + I_{\text{dark}} + \varepsilon_{\text{shot}}, \tag{6}$$

with $\text{Var}[\varepsilon_{\text{shot}}] \propto I_{\text{photo}}$; the proportionality coefficient is absorbed into per-profile calibration. The simulated photocurrent is renormalized per sample,

$$X_{\text{norm}} = \frac{I_{\text{photo}} - I_{\min}}{I_{\max} - I_{\min} + \epsilon}, \tag{7}$$

before dataset normalization.

[1]The absence of an explicit NL prefactor is intentional: this is a behavioral proxy for position-dependent update difficulty, not the strict derivative of a power-law $f(G) = G^{NL}$.

Table 2: Notation used for the Tiny-ViT experiment family in the main text.

ID	Meaning
V1	FP32 digital baseline
V2	Quantized hybrid control without analog noise
V3	Standard training with a fixed D2D realization; noisy deployment baseline
V4	Uniform-noise hardware-aware training result used as the canonical deployment model
V5	Pessimistic robustness stress test with larger uniform noise
V6	V4 plus physical front-end compensation
V7	Legacy retention-aware model, excluded from the canonical claims
V8	Corrected retention-aware follow-up used only for retention checks

5.4 Profile-Driven Substitution Interface

The profile interface packages technology-specific assumptions into a replaceable JSON parameter bundle. Deployment holds the digital backbone and training recipe fixed while substituting the profile, enabling literature priors today and measured-device calibration later without code changes.

5.5 Sensitivity and Energy Metrics

For the joint D2D–ADC sweep, first-order factor importance is quantified with Sobol indices

$$S_i = \frac{\mathrm{Var}_{X_i}(\mathbb{E}[Y \mid X_i])}{\mathrm{Var}(Y)}, \tag{8}$$

estimated from the 7×9 grid of Monte Carlo means. Energy is estimated via a first-order analytical model (Supplementary Methods) from operator counts and literature-proxy constants ($E_{\mathrm{cell}} = 100$ fJ, $E_{\mathrm{ADC},8b} = 25$ fJ, $E_{\mathrm{DAC},8b} = 30$ fJ); these are analytical placeholders for relative comparison, not measured silicon values.

6 Experimental Setup

We evaluate ResNet-18 on CIFAR-10 as an entry validation platform, ConvNeXt-Tiny trained from scratch as the higher-capacity CNN testbed, and Tiny-ViT-5M fine-tuned from ImageNet as the deployment-oriented transformer setting. CIFAR-10, CIFAR-100, and Flowers-102 provide a progression from easier to harder classification settings and from relatively data-rich to data-limited regimes.

The experiments fall into three groups: ResNet baseline quantization and front-end trends, ConvNeXt retention and proportional-noise studies, and Tiny-ViT hardware-aware training together with physical-extension tests such as nonlinear write and fresh-instance transfer. Detailed optimization settings and the Monte Carlo evaluation protocol are summarized in the Supplementary Information.

For ConvNeXt, C1–C4 denote the analogous FP32, quantized/no-noise, noisy-baseline, and hardware-aware training runs; C4 also anchors the proportional-noise and retention follow-up studies described in the Supplementary Information.

Parameter provenance is summarized in the Supplementary Information, together with the optimization settings and evaluation conditions used for the reported runs. The present study therefore targets reproducible reruns through archived configurations, logs, and model lineage. We also emphasize that the framework operates at the simulation level: all device parameters are literature-derived or proxy-calibrated rather than extracted from fabricated devices under direct measurement. Validation against physical hardware remains for future work, and the quantitative results below should be interpreted as comparative risk rankings rather than absolute deployment predictions.

7 Conclusion

We presented a first-order behavioral simulation framework for organic optoelectronic compute-in-memory inference that links materials-level characterization to task-level vision performance. By combining differential mapping, variability, retention, and peripheral constraints within a profile-driven workflow, the framework serves as a practical materials-to-system decision aid for evaluating deployment risk and comparing mitigation strategies before full array hardware becomes available.

Several observations stand out. A 63-point iso-accuracy sweep reveals a two-phase constraint hierarchy: ADC resolution below 6 bits causes abrupt collapse ($S_{\mathrm{ADC}} = 0.98$ over the full grid), while within the operational envelope, device-to-device variability becomes the binding factor ($S_{\mathrm{D2D}} = 0.92$). Fixed hardware-instance transfer, rather than nominal quantization, is the most prominent simulated deployment risk in the present regime: standard HAT collapses on fresh arrays, whereas Ensemble HAT raises fresh-instance accuracy from 10.00 % to 86.16±0.19% across three training seeds. In the canonical uniform-noise setting, quantization and nominal stochasticity are less limiting once scale recovery is active. By contrast, severe nonlinear write ($NL = 2.0$) recovers to the ∼80–82% band with the revised gradient-scaling recipe, leaving a residual gap of roughly 5 pp to the canonical $NL = 1.0$ baseline, and determining whether physical organic devices enter this regime will require direct nonlinear-write measurements.

The framework translates reported device behavior into deployment-facing system risk through an explicit profile interface. Under a literature-anchored OPECT case study, zero-shot simulated deployment reaches 88.53±0.08% ($n = 10$ fresh-instance evaluations) without any change to the evaluation workflow. The same structure should simplify future measured-device calibration while preserving model-level comparability within a common workflow.

Data Availability

All datasets used in this study (CIFAR-10, CIFAR-100, and Flowers-102) are publicly available through standard PyTorch torchvision and HuggingFace dataset interfaces. The source data underlying the main-text and Supplementary figures and tables are packaged in the submission-matched source-data archive together with the corresponding manifests, logs, and figure-generation inputs. A reproducibility archive containing the code snapshot, source data, SHA-256 manifest, and `CITATION.cff` file has been staged for Zenodo deposition and will be assigned a DOI at acceptance.

Code Availability

The simulation framework, training and evaluation scripts, device-profile utilities, and plotting code will be provided to editors and reviewers through a reviewer-accessible archive at submission. A curated public release will be made available at https://github.com/OrgOptEdge/compute_vit upon acceptance. The current repository is distributed under the Apache-2.0 licence, and the staged Zenodo archive will be updated with the final public DOI at acceptance.

Competing Interests

The authors declare no competing interests.

References

[1] Mark Horowitz. Computing's energy problem (and what we can do about it). In *IEEE International Solid-State Circuits Conference (ISSCC)*, pages 10–14, 2014. doi: 10.1109/ISSCC.2014.6757323. URL https://doi.org/10.1109/ISSCC.2014.6757323.

[2] Xiaochen Peng, Shanshi Huang, Hongwu Jiang, Anni Lu, and Shimeng Yu. DNN+NeuroSim v2.0: An end-to-end benchmarking framework for compute-in-memory accelerators for on-chip training. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 40(11):2306–2319, 2021. doi: 10.1109/TCAD.2020.3043731. URL https://doi.org/10.1109/TCAD.2020.3043731.

[3] Mario Lanza et al. The growing memristor industry. *Nature*, 640:613–622, 2025. doi: 10.1038/s41586-025-08733-5.

[4] Yunchao Xu, Yuan He, Dongyong Shan, Biao Zeng, and Qian-Xi Ni. Emerging artificial synaptic devices based on organic semiconductors: Molecular design, structure and applications. *ACS Applied Materials & Interfaces*, 2025. doi: 10.1021/acsami.4c17455. URL https://doi.org/10.1021/acsami.4c17455.

[5] Haotian Guo, Jing Guo, Yujing Wang, Hezhen Wang, Simin Cheng, Zehao Wang, Qian Miao, and Xiaomin Xu. An organic optoelectronic synapse with multilevel memory enabled by gate modulation. *ACS Applied Materials & Interfaces*, 2024. doi: 10.1021/acsami.3c19624. URL https://doi.org/10.1021/acsami.3c19624.

[6] Li Zhang, Wei Chen, and Jing Xu. Recent progress in organic optoelectronic synaptic devices. *Photonics*, 12(5):435, 2025. doi: 10.3390/photonics12050435. URL https://doi.org/10.3390/photonics12050435.

[7] Uihoon Jung, Miseong Kim, Jaewon Jang, Jin-Hyuk Bae, In Man Kang, and Sin-Hyung Lee. Formation of cluster-structured metallic filaments in organic memristors for wearable neuromorphic systems with bio-mimetic synaptic weight distributions. *Advanced Science*, 11 (9):2307494, 2024. doi: 10.1002/advs.202307494. URL https://doi.org/10.1002/advs.202307494.

[8] Jianmin Zeng, Xinhui Chen, Shuzhi Liu, Qilai Chen, and Gang Liu. Organic memristor with synaptic plasticity for neuromorphic computing applications. *Nanomaterials*, 13(5): 803, 2023. doi: 10.3390/nano13050803. URL https://doi.org/10.3390/nano13050803.

[9] Lan Shen Hu, Marco Fattori, Winston Schilp, Roy Verbeek, Setareh Kazemzadeh, Yoeri van de Burgt, Auke Jisk Kronemeijer, Gerwin Gelinck, and Eugenio Cantatore. An energy-efficient solid-state organic device array for neuromorphic computing. *IEEE Transactions on Electron Devices*, 70(12):6520–6525, 2023. doi: 10.1109/TED.2023.3327947. URL https://doi.org/10.1109/TED.2023.3327947.

[10] Xu Liu, Shilei Dai, Yiyang Jin, Junyao Zhang, Ziyi Guo, Tongrui Sun, Li Li, Pu Guo, Huaiyu Gao, Haixia Liang, Shiqi Zhang, Lize Xiong, Yanmin Zhou, and Jia Huang. Near-infrared organic photoelectrochemical synaptic transistors by wafer-scale photolithography for neuromorphic visual system. *Nature Communications*, 17:197, 2026. doi: 10.1038/s41467-025-66891-6. URL https://doi.org/10.1038/s41467-025-66891-6. Published online 2025-12-08.

[11] Dingyao Liu, Xinyu Tian, Jing Bai, Shaocong Wang, Shilei Dai, Yan Wang, Zhongrui Wang, and Shiming Zhang. A wearable in-sensor computing platform based on stretchable organic electrochemical transistors. *Nature Electronics*, 7:1176–1185, 2024. doi: 10.1038/s41928-024-01250-9.

[12] Padinhare Cholakkal Harikesh, Deyu Tu, and Simone Fabiano. Organic electrochemical neurons for neuromorphic perception. *Nature Electronics*, 7:525–536, 2024. doi: 10.1038/s41928-024-01200-5. URL https://doi.org/10.1038/s41928-024-01200-5.

[13] Yu-Pu Lin, Christopher H. Bennett, Théo Cabaret, Damir Vodenicarevic, Djaafar Chabi, Damien Querlioz, Bruno Jousselme, Vincent Derycke, and Jacques-Olivier Klein. Physical realization of a supervised learning system built with organic memristive synapses. *Scientific Reports*, 2016. doi: 10.1038/srep31932. URL https://doi.org/10.1038/srep31932.

[14] Vinay Joshi, Manuel Le Gallo, Simon Haefeli, Irem Boybat, S. R. Nandakumar, Christophe Piveteau, Martino Dazzi, Bipin Rajendran, Abu Sebastian, and Evangelos Eleftheriou. Accurate deep neural network inference using computational phase-change memory. *Nature Communications*, 11:2473, 2020. doi: 10.1038/s41467-020-16108-9. URL https://doi.org/10.1038/s41467-020-16108-9.

[15] Jungwook Choi, Zhuo Wang, Swagath Venkataramani, Pierce I-Jen Chuang, Vijayalakshmi Srinivasan, and Kailash Gopalakrishnan. PACT: Parameterized clipping activation for quantized neural networks. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1028–1037, 2018. doi: 10.48550/arXiv.1805.06085. URL http://proceedings.mlr.press/v80/choi18a.html.

[16] Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. In *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 23–30, 2017. doi: 10.1109/IROS.2017.8202133. URL https://doi.org/10.1109/IROS.2017.8202133.

[17] Pietro Belleri, Judith Pons i Tarrès, Iain McCulloch, Paul W. M. Blom, Zsolt M. Kovács-Vajna, Paschalis Gkoupidenis, and Fabrizio Torricelli. Unravelling the operation of organic artificial neurons for neuromorphic bioelectronics. *Nature Communications*, 15:5350, 2024. doi: 10.1038/s41467-024-49668-1. URL https://doi.org/10.1038/s41467-024-49668-1.

[18] Junpeng Ji, Dace Gao, Han-Yan Wu, Miao Xiong, Nevena Stajkovic, Claudia Latte Bovio, Chi-Yuan Yang, Francesca Santoro, Deyu Tu, and Simone Fabiano. Single-transistor organic electrochemical neurons. *Nature Communications*, 16:4334, 2025. doi: 10.1038/s41467-025-59587-4. URL https://doi.org/10.1038/s41467-025-59587-4.

[19] Fanqing Zhang, Xinyi Shan, Zhenhua Ji, Yan Wang, Jing Zhao, and Zhongrui Wang. Large-scale high uniform optoelectronic synapses array for artificial visual neural network. *Microsystems & Nanoengineering*, 11:42, 2025. doi: 10.1038/s41378-024-00859-2. URL https://doi.org/10.1038/s41378-024-00859-2.

[20] Heyi Huang, Xiangpeng Liang, Yuyan Wang, Jianshi Tang, Yuankun Li, Yiwei Du, Wen Sun, Jianing Zhang, Peng Yao, Xing Mou, Feng Xu, Jinzhi Zhang, Yuyao Lu, Zhengwu Liu, Jianlin Wang, Zhixing Jiang, Ruofei Hu, Ze Wang, Qingtian Zhang, Bin Gao, Xuedong Bai, Lu Fang, Qionghai Dai, Huaxiang Yin, He Qian, and Huaqiang Wu. Fully integrated multi-mode optoelectronic memristor array for diversified in-sensor computing. *Nature Nanotechnology*, 20(1):93–103, 2025. doi: 10.1038/s41565-024-01794-z. URL https://doi.org/10.1038/s41565-024-01794-z.

[21] Ying Liu, Biao Wang, Lei Wu, Lulu Huang, Lu Lin, Li Xiang, Dongqing Liu, Shiguo Zhang, Chenguang Zhu, and Yijie Tao. Artificial visual synaptic architecture with high-linearity light-modulated weight for optoelectronic neuromorphic computing. *ACS Applied Materials*

& Interfaces, 2023. doi: 10.1021/acsami.3c11495. URL https://doi.org/10.1021/acsami.3c11495.

[22] Dingwei Li, Yitong Chen, Huihui Ren, Yingjie Tang, Siyu Zhang, Yan Wang, Lixiang Xing, Qi Huang, Lei Meng, and Bowen Zhu. An active-matrix synaptic phototransistor array for in-sensor spectral processing. *Advanced Science*, 11, 2024. doi: 10.1002/advs.202406401. URL https://doi.org/10.1002/advs.202406401.

[23] Tianqing Wan, Bangjie Shao, Sijie Ma, Yue Zhou, Qiao Li, and Yang Chai. In-sensor computing: Materials, devices, and integration technologies. *Advanced Materials*, 35(37): 2203830, 2023. doi: 10.1002/adma.202203830. URL https://doi.org/10.1002/adma.202203830.

[24] A. Melianas, T. J. Quill, G. LeCroy, Y. Tuchman, H. v. Loo, S. T. Keene, A. Giovannitti, H. R. Lee, I. P. Maria, I. McCulloch, and A. Salleo. Temperature-resilient solid-state organic artificial synapses for neuromorphic computing. *Science Advances*, 6:eabb2958, 2020. doi: 10.1126/sciadv.abb2958. URL https://doi.org/10.1126/sciadv.abb2958.

[25] Ziyi Guo, Junyao Zhang, Ben Yang, Li Li, Xu Liu, Yutong Xu, Yue Wu, Pu Guo, Tongrui Sun, Shilei Dai, Haixia Liang, Jun Wang, Yidong Zou, Lize Xiong, and Jia Huang. Organic high-temperature synaptic phototransistors for energy-efficient neuromorphic computing. *Advanced Materials*, 36:2310155, 2024. doi: 10.1002/adma.202310155. URL https://doi.org/10.1002/adma.202310155.

[26] Corey Lammie, Wei Xiang, Bernabé Linares-Barranco, and Mostafa Rahimi Azghadi. Memtorch: An open-source simulation framework for memristive deep learning systems. *Neurocomputing*, 485:124–133, 2022. doi: 10.1016/j.neucom.2022.02.043. URL https://doi.org/10.1016/j.neucom.2022.02.043.

[27] Malte J. Rasch, Diego Moreda, Tayfun Gokmen, Manuel Le Gallo, Fabio Carta, Cindy Goldberg, Kaoutar El Maghraoui, Abu Sebastian, and Vijay Narayanan. A flexible and fast PyTorch toolkit for simulating training and inference on analog crossbar arrays, 2021. URL https://arxiv.org/abs/2104.02184.

[28] Sandia National Laboratories. Crosssim: Sandia's simulator for analog ai accelerators. Technical report, Sandia National Laboratories, 2024. URL https://doi.org/10.2172/2585829.

[29] Xueying Wu, Edward Hanson, Nansu Wang, Qilin Zheng, Xiaoxuan Yang, Huanrui Yang, Shiyu Li, Feng Cheng, Partha Pratim Pande, and Janardhan Rao Doppa. Block-wise mixed-precision quantization: Enabling high efficiency for practical ReRAM-based DNN accelerators. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2024. doi: 10.1109/TCAD.2024.3409193. URL https://doi.org/10.1109/TCAD.2024.3409193.

[30] Chenyu Wang, Zhen Dong, Daquan Zhou, Zhenhua Zhu, Yu Wang, Jiashi Feng, and Kurt Keutzer. EPIM: Efficient processing-in-memory accelerators based on epitome. In *Proceedings of the 61st ACM/IEEE Design Automation Conference (DAC)*, pages 1–6, 2024. doi: 10.1145/3649329.3657377. URL https://arxiv.org/abs/2311.07620.

[31] Mengke Ge, Junpeng Wang, Binhan Chen, Yingjian Zhong, Haitao Du, Song Chen, and Yi Kang. Allspark: Workload orchestration for visual transformers on processing in-memory systems. *IEEE Transactions on Computers*, 2025. doi: 10.1109/TC.2024.3483633. URL https://doi.org/10.1109/TC.2024.3483633.

[32] Cong Wang, Zexin Fu, Jiayi Huang, and Shanshi Huang. Hemlet: A heterogeneous compute-in-memory chiplet architecture for vision transformers with group-level parallelism, 2025. URL https://arxiv.org/abs/2511.15397.

[33] Shiwei Liu, Chen Mu, Hao Jiang, Yunzhengmao Wang, Jinshan Zhang, Feng Lin, Keji Zhou, Qi Liu, and Chixiao Chen. HARDSEA: Hybrid analog-ReRAM clustering and digital-SRAM in-memory computing accelerator for dynamic sparse self-attention in transformer. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 32(2):269–282, 2024. doi: 10.1109/TVLSI.2023.3337777. URL https://doi.org/10.1109/TVLSI.2023.3337777.

[34] W. Wan, R. Kubendran, B. Gao, S. Joshi, P. Raina, H. Wu, G. Cauwenberghs, and H.-S. P. Wong. A voltage-mode sensing scheme with differential-row weight mapping for energy-efficient RRAM-based in-memory computing. In *2020 IEEE Symposium on VLSI Technology*, pages 1–2, 2020. doi: 10.1109/VLSITechnology18217.2020.9265066. URL https://doi.org/10.1109/VLSITechnology18217.2020.9265066.

[35] Xianggao Wang, Na Wei, Shifan Gao, Wenhao Wu, and Yi Zhao. Accuracy improvement with weight mapping strategy and output transformation for STT-MRAM-based computing-in-memory. *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits*, 10:75–81, 2024. doi: 10.1109/JXCDC.2024.3478360. URL https://doi.org/10.1109/JXCDC.2024.3478360.

[36] Tomas Vincze, Michal Hanic, Martin Berki, and Martin Weis. Simultaneous dual-plasticity organic synaptic transistor for neuromorphic computing. *Advanced Electronic Materials*, 12(1):e00515, 2026. doi: 10.1002/aelm.202500515. URL https://doi.org/10.1002/aelm.202500515. Published online 2025-12-03.